Do Now · retrieval
Hexadecimal, shifts & masks

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Choose a type

Q2

Read an unsigned byte

Q3

Encode a negative number

Q4

Add and check the range

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Hexadecimal links directly to binary — Each hexadecimal digit maps to exactly four bits because four bits have 16 possible patterns. — CODE:
$6D_{16}$
6 → 0110
D → 1101
= 01101101_2

Core idea 2

Logical shifts move bits — A logical shift inserts zeros into the empty positions. Bits leaving the fixed width are lost.

Core idea 3

AND and OR masks — AND: mask 1 preserves a bit; mask 0 clears it. — OR: mask 1 sets a bit; mask 0 preserves it.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

XOR toggles selected bits — XOR produces 1 when the two input bits differ. A mask bit of 1 flips the corresponding data bit; 0 leaves it unchanged.

Vocabulary · use the correct technical terms

Hexadecimal — Base 16, using digits 0–9 and letters A–F.

Nibble — A group of four bits.

Logical shift — Moving bits left or right, filling empty positions with zeros.

Mask — A bit pattern combined with another using a bitwise operation.

Independent skill check · apply the lesson

Starting with 10100101, calculate (a) a logical shift right by 2 and (b) AND with 00001111. Explain each result.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Link denary, hexadecimal and binary

Follow the worked method

1. Find the hex digits — CODE:
173 ÷ 16 = 10 remainder 13 — The quotient gives the first hex digit; the remainder gives the second.

2. Use four bits per hex digit — CODE:
A → 1010
D → 1101

$AD_{16} = 10101101_2$

3. Return to denary — CODE:
(10 × 16) + 13 = 173

128 + 32 + 8 + 4 + 1 = 173 — For binary → hex, group bits from the right in fours and pad the left group with zeros if needed.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the answer

CLUES · uncover one at a time

Clue 1: Three hexadecimal digits have place values 256, 16 and 1.

Clue 2: The hexadecimal digit F represents denary 15.

Clue 3: Multiply each digit by its place value before adding.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: Write the three place values.

Step 2: Calculate each weighted contribution.

Step 3: Add to give the denary result.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Hexadecimal → binary

Write the binary equivalent of hexadecimal B7E .

Paraphrased exam question; original value and mark retained. OCR H446/01, page 17 .

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Control an eight-bit register
Bits are numbered 7…0 from left to right. Give an operation and an eight-bit mask for each independent task. Preserve all bits not mentioned.
Clear bits 7 and 6. [2]
Set bits 3 and 0. [2]
Toggle bits 5 and 2. [2]
Original exam-style application; one mark for the operation and one for the mask.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

One hexadecimal digit represents four binary bits.

Use key vocabulary from memory

1. Explain Hexadecimal in your own words.

2. Explain Nibble in your own words.

3. Explain Logical shift in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

2F3 = (2 x 256) + (15 x 16) + (3 x 1) = 755.

PAGE 3 · Independent skill check:

Logical >>2 gives 00101001 (discard two rightmost bits, fill with zeros). AND 00001111 gives 00000101 (clear upper four bits).

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-003/answers.html